

Project-based Learning

Project II

Briefing

Project II (M-CCIA)
Semester II Units (M-CCIA)

DEGREE: Masters in Creative Computing and Artificial Intelligence (M-CCIA)

UNIT: Project II

YEAR/SEMESTER: 2025-2026 / 2nd semester

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UC: Produção de Moda / 45 HRS / 5 ECTS

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1. INTRODUCTION

This document presents the semester project developed under a Project-Based Learning (PBL) methodology. The project brings together contributions from four supporting curricular units—Generative AI, Applied Deep Learning, Interface Design and Interaction, and Emerging Technologies—coordinated through Project II as the integrating and management framework. Together, these courses support the conception, design, and development of an **interactive Emergent Technology experience**.

The overarching goal is for students to produce a fully functional **interactive prototype** that demonstrates both technical competence and creative intention. Students are expected to combine methods and practices from Generative AI, Applied Deep Learning, Emerging Technologies, Interface Design and Interaction, and project planning and production within Project II, in order to develop an experience that is coherent, meaningful, and engaging for a public audience.

Each course contributes essential knowledge and skills to the project's success:

- **Generative AI** introduces prompt-based workflows and creative uses of generative systems to support ideation, concept development, and asset production (e.g., text, images, sound, or video), as well as critical reflection on authorship, ethics, and limitations.
- **Applied Deep Learning** supports the understanding and practical application of generative models (e.g., CNN-based approaches, GANs, VAEs, diffusion models) for creating, transforming, or driving content and behaviours within the experience.
- **Emerging Technologies** explores VR/AR ecosystems and sensor-based interaction, including body/hand tracking, spatial input, and relevant hardware/software frameworks, enabling students to integrate modalities such as hand tracking, eye tracking, or haptic feedback where appropriate.
- **Interface Design and Interaction** focuses on user experience and interaction quality, ensuring that interactions are intuitive, comfortable, and accessible in immersive contexts. This includes affordances, feedback, usability testing, and iterative refinement to support immersion and legibility.
- **Project II** provides the integration and project management layer, guiding students through planning, prototyping, production, documentation, and iteration. It ensures that the contributions from the other courses converge into a cohesive final artefact.

By the end of the semester, students are expected to have:

- Developed a working Emergent Technology product that meaningfully integrates interaction design, emerging technologies, and generative AI/deep learning components.
- Gained hands-on experience with Unity and immersive development workflows.
- Conducted usability testing and iterative improvements informed by observation and feedback.
- Strengthened collaboration, communication, and project management skills through interdisciplinary teamwork.
- Prepared the prototype for a public exhibition, including presentation materials and a clearly communicated concept.

This document outlines the project theme, technical expectations, milestones, and evaluation criteria to support a structured development process and ensure readiness for the end-of-semester public showcase.

2. THEME

The results of these projects will be integrated into the fashion event calendar organized by the **Master's in Branding and Fashion Design**, who will be responsible for the event's production and curation.

Students from the **MCCIA (Master in Creative Computing and Artificial Intelligence)** are expected to develop an exploratory and speculative project, suitable for public presentation as an interactive exhibition experience.

The theme and output format are open, but must relate to the fashion universe in some way. Examples of potential topics to address include:

- **Identity:** In what ways does our expression of identity evolve with technology?
- **Body:** How does technology become a bodily extension, changing the way the body is perceived and presented?
- **Society:** How will technology impact how we build community, relate to one another, and communicate through what we wear?
- **Culture:** How will new technologies impact the communication of cultural concepts and messages?
- **Wearables:** In what ways will technology be used functionally and/or communicatively on the body?
- **Industry 5.0:** How will humans and intelligent machines collaborate in creative and productive processes?
- **Consumption:** How will technology alter the shopping experience for fashion products?

In the first phase, students must decide the topic they would like to speculate about, find information that supports it (critically and theoretically) and define what kind of technologies they would like to work with.

3. Project Phases

Phase 1: Research and Concept Development (Weeks 1-3)

In this phase, students should:

- Research possible methods of Emergent Technologies, selecting one that will serve as a reference for the development of the experience - each group should try to select a different technology.
- Present references for style, image, color, sound, etc. for experience.
- Present the user flow of the experience, what does he do, how does he interact with, what are the outcomes, etc...

Phase 2: Prototyping and Technical Exploration (Weeks 3-7)

In this phase, students should:

- Create a list of tasks required to be completed to have the functional project at the end of the semester;
- Identify the problems that might arise, as well as the necessities of the project;

- Develop a function prototype that demonstrates the functionalities of the product

Phase 3: Refinement and User Experience (Weeks 8-11)

In this phase, students should:

- Refine the project to a final product state;
- Test the project with other users to identify problems from not knowing use

Phase 4: Presentation and Exhibition (Weeks 12-14)

In this phase, students should:

- Prepare the project for the final presentation;
- Identify special needs required for the presentation and complete the fulfillment of those needs;

4. PROJECT REQUIREMENTS

Within the context of the collaboration between the two master programs, the groups having the opportunity propose solutions and strategies for any undefined aspects of their project, but must comply with the following requirements:

- The team size is **2 students** (exceptions only if needed).
- The prototype must be testable.
- **Graphical and audio assets** must be integrated.
- All assets must be **original or legally obtained** from 3rd parties.

Strategic decisions should be presented to the project professors as soon as possible for the project to be accepted and some initial feedback given.

5. COMPLEMENTARY CURRICULAR UNIT CONTRIBUTIONS

ECTS	HRS /WEEKS	COMPLEMENTARY CURRICULAR UNIT CONTRIBUTIONS TO THE PROJECT	PROFESSOR/email	% PROJECT
5	2 h 14 w	INTERFACE DESIGN AND INTERACTION Human-computer interaction design.	Paula Neves paula.neves@univeridadeeuropeia.pt	60%
5	2 h 14 w	GENERATIVE AI Prompt engineering for text, image and sound models. Prompt development for generating new content, task execution, and Agentic AI.	Cláudia Sevivas claudia.sevivas@univeridadeeuropeia.pt	50%
5	2 h 14 w	APPLIED DEEP LEARNING Generative models for text, image, and sound. Model training and fine-tuning. Word embedding and Latent space manipulation for creative outputs.	Cláudia Sevivas claudia.sevivas@univeridadeeuropeia.pt	50%
5	2 h 14 w	EMERGING TECHNOLOGIES VR/AR interaction design.	Fernando Marson fernando.marson@univeridadeeuropeia.pt	50%

6. PROJECT MILESTONES

Continuous assessment has three milestones, each requiring a minimum of **70% assiduity** (refer to the regulation for special cases). **Exam assessment** has a single milestone. Each milestone consists of a **delivery** made through Canvas, a **presentation** for teachers and students, and invited guests for feedback.

■ 1st milestone: Documentation (MCCIA)

- **Presentation:** 2026/02/16
- **Delivery:** 2026/02/16, 23:59:59 GMT
- **Deliverables**
 - Presentation slides.
 - Production dossier/report (identifies the participation of the “big 4” units), including:
 - Weekly activity plan, contextualized by each unit.
 - Research about related projects
 - Emerging Technologies: Select the most appropriate hardware and frameworks for the project and define a clear integration strategy, outlining how these technologies will enhance interactivity and immersion within their digital fashion experience.

■ 2nd milestone: Prototype (MCCIA)

- **Presentation:** 2026/04/13
- **Delivery:** 2026/04/12, 23:59:59 GMT
- **Deliverables**
 - Presentation slides
 - Production dossier/report (contains chapters about all the “big 4” units), including:
 - Activity plan execution record and update.
 - Workload distribution between group members.
 - Emerging Technologies: Report about the implementation and test of sensor-based interactions, integrating functionalities (i.e. body tracking, gesture recognition, and AR) within the project.
 - Applied Deep Learning: At this stage, the project must demonstrate a functional prototype that integrates generative models for text, image, and/or sound into the interactive experience. Students are expected to implement and test at least one trained or fine-tuned model (e.g., VAE, GAN, diffusion-based or embedding-driven system) that meaningfully contributes to the concept. This includes documenting the training or fine-tuning process, dataset selection, parameter decisions, and initial experiments with word embeddings or latent space manipulation to generate controlled creative outputs. The prototype should clearly show how generative processes are embedded within the interactive flow, rather than used merely as static asset generators. TouchDesigner should be used as one of the core tools to prototype real-time generative behaviours, visual outputs, or audiovisual integration, particularly where latent space exploration or model-driven transformations are involved. However, students may integrate additional tools (e.g., Python/TensorFlow, PyTorch, Unity, ONNX workflows, or external APIs) where necessary to support training, inference, or multimodal experimentation. The emphasis at this milestone is on technical feasibility, conceptual coherence, and the first validation of generative systems within an interactive framework.

- Generative AI: By this milestone, the team should deliver a working prototype that demonstrates the project's core interactive idea and proves the technical feasibility of the experience, while explicitly integrating prompt engineering workflows for text, image and sound to produce or transform key assets and behaviours. The prototype should include at least one meaningful pipeline where prompts are used not only for "content generation" (e.g., text copy, image prompts, sound prompts) but also for task execution (e.g., structured outputs, content variations, system/prompt templates) and an initial agentic loop (e.g., plan → generate → evaluate → iterate) documented in the report. At minimum, students should integrate TouchDesigner as a real-time system in the prototype (e.g., driving visuals, reactive UI, or mapping generated text/sound parameters to visuals), while allowing complementary tools if needed for image or sound generation—provided the prompt rationale and iteration process are clearly described in the production dossier.
- Project Prototype
- All the source code

■ 3rd milestone: Final product (MCCIA)

- **Presentation:** 2025/06/05
- **Delivery:** 2025/05/29, 23:59:59 GMT
- **Deliverables** (same for exams)
 - Presentation Slides
 - Production dossier/report (contains chapters about all the "big 4" units), including the same as the 2nd milestone, and:
 - Reflection on the planning effort and work distribution.
 - Emerging Technologies: Discuss how the work can contribute to fashion innovation in modern technologies.
 - Applied Deep Learning: In the final milestone, students must refine and stabilise the integration of generative models within the complete interactive experience. The project should demonstrate deliberate artistic control over model behaviour through latent space manipulation, embedding navigation, conditioning strategies, or fine-tuned parameters. Generative systems must move beyond experimentation to become structurally embedded in the user experience, shaping text, image, or sound outputs in response to interaction, sensor data, or system logic. TouchDesigner should be fully integrated where appropriate to support real-time visualisation, generative pipelines, and multimodal interaction. Additional tools may continue to be used for model optimisation, deployment, or cross-platform integration. The final report must reflect critically on model performance, creative affordances of the latent space, limitations encountered during training or inference, and how generative deep learning meaningfully contributed to the speculative and experiential goals of the project.
 - Generative AI: By the final milestone, the team should deliver a refined, exhibition-ready interactive experience where generative AI is not an add-on but a coherent creative mechanism supporting the concept and user journey

(including stability, responsiveness, and clarity of interaction). Prompt engineering should be treated as a designed system: reusable prompt templates, constraints, style guides/personas, safety/quality checks to reduce hallucinations or inconsistent outputs, and (where relevant) a more mature agentic workflow that reliably executes tasks (e.g., generating variations, selecting outputs, logging decisions, or adapting content to user input). TouchDesigner should be integrated in the final build as a polished real-time component (e.g., final visual system, reactive mapping, audiovisual orchestration), and the dossier should document how prompting decisions shaped the final aesthetic/interaction outcomes, alongside iterative improvements and performance/robustness considerations expected at the “final product state”.

- Project Prototype
 - Emerging Technologies: Optimize and refine interactive technologies, improving accuracy, responsiveness, and performance.
- All the source code

6. PROJECT ASSESSMENT

Each milestone contributes to the final grade:

Continuous Evaluation

- **1st milestone:** 15% of the grade
- **2nd milestone:** 35% of the grade
- **3rd milestone:** 50% of the grade

Final Evaluation

- **Exam milestone:** 100% of the grade